
Symmetries in Physics — Exercise Sheet 2

Structure of groups, quotients, and products

Variant: Questions only

Exercise 1. (i) Show that every subgroup $H \leq G$ of index 2 is normal. (ii) Deduce the following. Let a finite group $G \leq O(n)$ contain an element of determinant -1 (for $n = 2$, a reflection). Then the subgroup of determinant-1 elements — the *rotation subgroup* — is normal of index 2.

Exercise 2. Compute the conjugacy classes of (i) D_4 and (ii) S_4 , and verify the class equation $|G| = \sum_i |C_i|$ in both cases.

Exercise 3. Use the first isomorphism theorem to identify the following quotients: (i) S_n/A_n for $n \geq 2$; (ii) $GL(n, \mathbb{R})/SL(n, \mathbb{R})$; (iii) $\mathbb{R}/2\pi\mathbb{Z}$.

Exercise 4. Show that $\mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}/mn\mathbb{Z}$ if and only if $\gcd(m, n) = 1$. The direct product is defined in the reading.

Exercise 5. Find the centre $Z(D_4)$, and show that the quotient $D_4/Z(D_4)$ is isomorphic to the Klein four-group.

Exercise 6 (semidirect products). Let $N \trianglelefteq G$ and $H \leq G$ satisfy $N \cap H = \{e\}$ and $NH = G$. (i) Show every $g \in G$ is uniquely $g = nh$ ($n \in N, h \in H$), and that the product reads

$$(n_1 h_1)(n_2 h_2) = (n_1 h_1 n_2 h_1^{-1})(h_1 h_2)$$

— the (*internal*) *semidirect product* $N \rtimes H$, with H acting on N by conjugation. (ii) Exhibit $D_n \cong C_n \rtimes C_2$ and identify the action of C_2 on C_n .

Exercise 7 (composition series). A *composition series* of a finite group G is a chain $\{e\} = G_0 \trianglelefteq G_1 \trianglelefteq \cdots \trianglelefteq G_k = G$ in which each quotient G_{i+1}/G_i is simple. Lecture 2 writes the same chain from G downwards; here it is read from $\{e\}$ upwards. (i) Exhibit a composition series for D_4 and list its factors. (ii) Find a *second*, genuinely different series for D_4 and check it has the same factor multiset — the Jordan–Hölder theorem in miniature.